

ESP32 Campus Security Gateway: Complete Web Portal Guide

Comprehensive Reference for Web Dashboard Features, Form Inputs, Tooltips & Administration

1. Access & Authentication

1.1 Connecting to the Configuration Hotspot

1. Place the ESP32 into **Setup Mode** by holding the physical Mode Button (GPIO 34) for 3 seconds or by sending an SMS with `AP ON` to the device.
2. The Blue Status LED on the unit will begin **blinking rapidly** (400ms interval).
3. Connect your smartphone, tablet, or laptop to the broadcasted Wi-Fi network:
 - **Default SSID:** `AntiWallJump-Setup`
 - **Default Wi-Fi Password:** `configure123`
4. The captive portal login screen will launch automatically. If it does not, open any browser and navigate to `http://192.168.4.1`.

1.2 Web Authentication

- Enter the default administrative password: `admin`
- Click **Log In** to receive an authenticated Bearer token stored in your browser session.

2. Persistent Header Bar & Global Actions

The top navigation header remains visible across all pages and provides real-time system status and quick-access controls.

2.1 Real-Time Status Badges

BADGE NAME	HTML IDENTIFIER	SOURCE	DESCRIPTION
System Clock	<code>#hdrTime</code>	DS3231 RTC	Displays the live local time formatted as <code>HH:MM:SS AM/PM</code> . Updates every second.
Node Identifier	<code>#hdrNode</code>	<code>sysConfig.nodeID</code>	Shows the configured alphanumeric name of the alarm unit (e.g., <code>NORTH_WALL_01</code>).
Cellular Signal	<code>#hdrSimStatus</code>	SIM800L Modem	Displays GSM network registration status: Registered (Green) when online, or Searching / No SIM (Red) when offline.
SIM Number	<code>#hdrSimNumber</code>	SIM800L Modem	Displays the mobile subscriber number of the installed SIM card.

2.2 Global Control Buttons

- **Sync Clock** (): Immediately reads your device's local Unix timestamp and updates the hardware DS3231 RTC chip.
- **Theme Switcher** (): Toggles between Dark Mode and Light Mode. Your preference is saved in local browser storage.
- **Logout** (): Clears the active authentication session and redirects back to the login screen.
- **Stop Portal** (): Displays a confirmation dialog, shuts down the Wi-Fi Access Point, re-arms the perimeter tripwire, and returns the device to Normal Guarding Mode.

3. Tab 1: Dashboard (#page-dashboard)

The Dashboard provides live visibility into perimeter security status, hardware health, and recent infraction records.

3.1 Security Status Banner

- **SYSTEM SECURED** (Green): The infrared beam is intact and the system is actively guarding the perimeter.
- **SECURITY COMPROMISED - ALARM ACTIVE** (Pulsing Red): The infrared beam tripwire has been interrupted and an active alarm/bucketing cycle is running.

3.2 "Test SMS" Button (#btnTestSms)

- Clicking **Test SMS** sends an immediate test text message to the configured emergency phone number.
- Verifies SIM prepaid balance, antenna signal strength, and network dispatch without tripping the physical alarm.

3.3 Node Overview Stat Cards

CARD LABEL	HTML ID	DATA SOURCE	TOOLTIP & EXPLANATION
Node ID	#statNodeID	sysConfig.nodeID	Tooltip: "The unique name and software version of this alarm unit." Identifies which physical security unit is reporting across campus.
Storage Media	#statSDStatus	SPI SD Driver	Tooltip: "Shows if the memory card is working to save logs." Displays Mounted & Active (Green) or Missing / Failed (Red).
Hardware Clock	#statTime	DS3231 RTC	Tooltip: "The current time, automatically updated over Wi-Fi." Displays the high-precision hardware time used for all audit logging.
Alert Recipient	#statPhone & #statRole	guardPhone	Tooltip: "The phone number that will receive the text messages." Displays the active emergency contact number and recipient role (GUARD / TEACHER).
Sensor Config	#statSensors	System Core	Tooltip: "The physical setup of the infrared beam sensors." Confirms active sensor topology (Active IR Beam).
Audio Module	#statAudio	UART1 Driver	Tooltip: "Shows if the alarm speaker is working properly." Displays Ready (Green) or Missing / Error (Red) for the DFPlayer Mini MP3 decoder.
Zone Assigned	#statWall	wallLabel	Tooltip: "The name of the wall or fence being protected." Short boundary sector label (e.g., North Wall Fence).
Exact Location	#statLocation	locationDesc	Tooltip: "The specific location description for the guards." Detailed physical description to guide security personnel to the exact site.
Sensor Status	#statSensor	GPIO 14 / Wiegand	Tooltip: "Shows if the jump sensors and ID scanner are active." Live dual indicator: • Port 14 (IR Beam): CLEAR (Green) or TRIGGERED (Red) • D0/D1 (RFID): STANDBY or READING... (Blue)

3.4 Immediate Incident Flag Table

- Displays the most recent security infractions registered during the current run cycle (**Timestamp** , **Status** , **Entity Name** , **Tag ID**).

4. Tab 2: Settings (#page-settings)

The Settings tab contains three sub-sections. Any modified field automatically turns the respective **Save Settings** button orange with an asterisk (Save Changes *) until saved.

4.1 Sub-Tab: Device Config (#cfg-device)

Controls hardware identification, emergency contact routing, and SMS permission filters.

FIELD LABEL	HTML ID	DEFAULT	CONSTRAINTS	TOOLTIP & EXPLANATION
Device ID	#cfgNodeID	NODE_01	1–31 Characters	Tooltip: "Name of this alarm device (e.g., Gate 1)." Unique alphanumeric identifier for this security unit.
Wall Label	#cfgWallLabel	Wall Section	1–31 Characters	Tooltip: "Name of the physical wall or boundary being protected (e.g., Section A)." Primary location headline placed inside SMS alert text messages.
Location Description	#cfgLocationDesc	Campus Perimeter	1–127 Characters	Tooltip: "A clear description of where this wall is located to help guards find it." Full geographic description (e.g., "Behind Chemistry Laboratory Complex Building B").
Emergency Contact Number	#cfgPhone	+639XXXXXXXX	Max 31 Characters	Tooltip: "The phone number that will receive the text message alerts." Supports standard local formats (09XXXXXXXX), international formats (+639XXXXXXXX), or 10-digit mobile numbers (9XXXXXXXX). Automatically normalized on save.
Number Lock	#cfgNumberLock	OFF (0)	Toggle (ON / OFF)	Tooltip: "If checked, only the emergency phone number is allowed to send text commands to the alarm." When enabled, incoming SMS commands (AP ON , AP OFF) are ignored unless sent from the configured emergency phone number.
Recipient Role	#cfgRecipientRole	Security Guard	Guard / Teacher	Tooltip: "Choose the recipient role for SMS alerts (e.g., Guard or Teacher)." Sets the SMS header: [CAMPUS ALERT - SECURITY] for Guard, or [CAMPUS ALERT - FACULTY] for Teacher.

4.2 Sub-Tab: Alert Triggers & Timers (#cfg-alert)

Controls the FreeRTOS event collection engine, audio siren playback, and sensor hold windows.

FIELD LABEL	HTML ID	DEFAULT	RANGE	TOOLTIP & EXPLANATION
Alarm Duration (ms)	#cfgDuration	30000	1,000–300,000 ms	Tooltip: "How long the alarm siren will sound when someone jumps the wall. (e.g., 30000 = 30 seconds)." Controls the runtime of the DFPlayer Mini siren and the Red Alarm LED (GPIO 4).
Cooldown Time (ms)	#cfgCooldown	60000	5,000–600,000 ms	Tooltip: "Wait time after an alarm before the system can trigger again. Prevents SMS and siren spam. (e.g., 60000 = 60 seconds)." Mandatory quiet period after an alarm to avoid redundant alerts while guards respond.
Breach Memory (ms)	#cfgBreachMemoryMs	5000	1,000–60,000 ms	Tooltip: "How long the system holds a motion sensor trigger before clearing it. (e.g., 5000 = 5 seconds)." Duration a beam-break pulse is remembered in memory while waiting to pair with an RFID card read.
RFID Memory (ms)	#cfgTagMemoryMs	30000	1,000–600,000 ms	Tooltip: "Time window a student has to cross the beam after scanning their ID. If they cross within this time, they are caught. (e.g., 30000 = 30 seconds)." Retention window for buffered RFID cards approaching the perimeter.
Bucketing Window (ms)	#cfgBucketWindowMs	10000	1,000–60,000 ms	Tooltip: "Time window after an alarm triggers to group other jumping students into the same SMS alert. (e.g., 10000 = 10 seconds)." Aggregation window to group accomplices jumping together into a single SMS report.
Unsent SMS Expiry (min)	#cfgSmsAlertExpiryMinutes	60	1–1,440 min	Tooltip: "Failed SMS alerts are retried until this time limit is reached. Valid range: 1 to 1440 minutes." Time queued SMS alerts will be retried during cellular network outages before expiration.
Beam Alert Delay (min)	#cfgPersistentBeamAlertMinutes	15	1–1,440 min	Tooltip: "Send one SMS when the IR beam stays broken for this long. Another SMS is allowed only after the beam recovers. Valid range: 1 to 1440 minutes." Dispatches a persistent obstruction warning if the IR beam remains continuously broken.

Audio Track Controls & Voice Testing

- **Alarm Voice Track Selector (#cfgAlarmVoice):** Selects from 8 MP3 tracks stored on the DFPlayer Mini SD card (/mp3/0001.mp3 through /mp3/0008.mp3).
- *Track 1:* "Please return to class..."
- *Track 2:* "Security alert. You are in a restricted zone..."
- *Track 3:* "Warning. Unauthorized access detected..."
- *Track 4:* "Attention. Climbing the fence is prohibited..."
- *Track 5:* "This is a restricted area. CCTV recording in progress..."
- *Track 6:* "Emergency alert. Perimeter breach detected..."
- *Track 7:* High-Decibel Siren 1
- *Track 8:* High-Decibel Siren 2
- **Play Button:** Auditions the currently selected track through the connected speaker.
- **Stop Button:** Immediately halts audio playback.
- **System Voice Prompt Tests:**
 - *Save Conf:* Plays Track 12 ("Configuration Saved").
 - *Load Avail:* Plays Track 13 ("SIM Load Available").
 - *No Load:* Plays Track 14 ("SIM Card Insufficient Balance").

4.3 Sub-Tab: System Security (#cfg-security)

Controls Wi-Fi credentials, gateway IP addressing, administrative passwords, and system resets.

FIELD LABEL	HTML ID	DEFAULT	CONSTRAINTS	TOOLTIP & EXPLANATION
New AP Name	#cfgApSSID	AntiWallJump-Setup	1–31 Characters	Tooltip: "The new Wi-Fi name this alarm will broadcast." Configures the SoftAP network SSID broadcast during configuration mode.
New AP Password	#cfgApPassword	configure123	8–63 Characters	Tooltip: "The new password to connect to this alarm's Wi-Fi." WPA2 Wi-Fi passphrase. Must be at least 8 characters.
Confirm AP Password	#cfgApPasswordConfirm	—	Must match AP Pass	Tooltip: "Type the new Wi-Fi password again to confirm." Verifies no typos were entered in the AP password field.
Device IP Address	#cfgDeviceIP	192.168.4.1	Valid IPv4	Tooltip: "The web address used to access this page. Leave as default unless needed." Gateway IP address of the Access Point.
New Admin Password	#cfgAdminPassword	admin	Min 4 Characters	Tooltip: "The new password to log into this webpage." Sets the password required to log into the web dashboard.
Confirm Admin Password	#cfgAdminPasswordConfirm	—	Must match Admin Pass	Tooltip: "Type the new webpage password again to confirm." Verifies no typos were entered in the admin password field.
Reboot Hardware	/api/reboot	—	Action Button	Safely reboots the ESP32 microcontroller without deleting any saved configurations or logs.
Factory Reset	/api/factory_reset	—	Action Button	Opens a confirmation modal requiring the user to type CONFIRM . Wipes EEPROM, deletes SD card logs and student registries, restores factory defaults, and restarts.

5. Tab 3: Student Registry (#page-registry)

The Registry tab is used to enroll authorized student RFID cards, link them to student names and ID numbers, and review individual violation histories.

5.1 Enrolling a Student Card (Step-by-Step)

1. Click **Scan Card** (#btnScan). The scanner enters a 30-second live listening mode with an animated spinner.
2. Tap or hold the student's UHF RFID card in front of the RD906M panel:
 - **Success:** The reader decodes the tag and auto-populates the **RFID Tag ID** field (#regTagID).
 - **Collision Warning:** If multiple cards are detected simultaneously, the system displays: " ⚠ Multiple cards detected! Clear the area and scan only ONE card."
 - **Duplicate Tag Warning:** If the card is already in the database, the system identifies the currently assigned student name.
3. Enter the student's **Complete Name** in #regName (Tooltip: "The full name of the student or staff member.").
4. Enter the student's **Official Student ID** in #regStudentID (Tooltip: "The official student ID number.").
5. Click **Register Student**. The record is saved to /students.csv on the MicroSD card.
 - If a conflict occurs, a confirmation dialog offers to reassign the tag or update the existing student profile.

5.2 Registered Students Table

TABLE COLUMN	DATA CONTENT	FUNCTIONAL ACTION
TAG ID	Hexadecimal card ID (e.g., 1A2B3C4D)	Unique hardware identifier.
COMPLETE NAME	Student Full Name (e.g., Juan DeLaCruz)	Used in SMS alerts and event logs.
OFFICIAL STUDENT ID	Institutional ID (e.g., 2026-0413)	Clickable blue link that opens the Student Incident History Modal .
FREQUENCY	Total breach count (e.g., 3)	Highlighted in red if the student has triggered the alarm.
ACTION	Drop button	Displays a confirmation dialog to permanently remove the student record.

5.3 Student Incident History Modal (#history-modal)

- Clicking any student's blue ID number opens their dedicated violation log parsed directly from /logs/<tagID>.csv .
- Displays an infraction table containing:
- **Timestamp:** Exact RTC date and time of the violation.
- **Wall:** Boundary zone breached.
- **Severity:** Infraction classification badge (CRITICAL or WARNING).

6. Tab 4: Records & Data Logging (#page-logs)

The Records tab provides access to master event logs and data export tools.

6.1 Event Logs Table

- Chronological list of all perimeter incidents parsed from `/eventlog.csv`.
- Columns:
- **TIMESTAMP:** Date and time of incident (e.g., `2026-08-17 14:30:15`).
- **STATUS:** Severity badge: `CRITICAL` (Red) for student breaches and unidentified trips, or `WARNING` (Yellow) for unregistered cards.
- **NAME:** Full student name, or `Unregistered` / `Unidentified`.
- **TAG ID:** Hexadecimal card ID, or `NONE`.

6.2 Data Management Buttons

- **Download Logs (#downloadLogs):**
- Generates and downloads a multi-sheet spreadsheet named `Campus_Security_Logs.csv`.
- Contains three distinct sections:
 1. *Master Chronological Event Log*
 2. *Student Registry Summary & Infraction Counts*
 3. *Individual Student Incident Breakdowns*
- **Mobile Browser Note:** If your phone's captive portal browser suppresses downloads, open Google Chrome or Apple Safari, type `http://192.168.4.1`, and click Download Logs.
- **Clear Logs (#clearLogs):**
- Displays a confirmation prompt before permanently deleting all event logs from the MicroSD card.

7. Web Portal Troubleshooting Matrix

ISSUE	PROBABLE CAUSE	CORRECTIVE ACTION
Cannot open 192.168.4.1	Wi-Fi disconnected or incorrect IP address.	Verify Wi-Fi is connected to AntiWallJump-Setup . Ensure mobile data is temporarily disabled if your phone tries to route traffic over cellular.
Login fails with "Unauthorized"	Incorrect password entered.	Default password is admin . If changed and forgotten, send SMS command RESTORE FACTORY SETTINGS from the Guard phone to reset.
Save Settings shows "Invalid phone number"	Number does not match carrier rules.	Enter a valid 11-digit local (09123456789) or 13-digit E.164 (+639123456789) number.
Scan Card times out after 30 seconds	Reader disconnected, unpowered, or card out of range.	Ensure RD906M reader has 12V DC power and Wiegand D0/D1 are connected to GPIO 21 and 22.
Log download does not start on mobile	Captive portal browser sandbox restriction.	Open the native mobile browser (Chrome/Safari), enter http://192.168.4.1 , log in, and tap Download Logs.